SMITH ENGINEERING
Queen's University

Mechatronics and Robotics Engineering

Calculation Sheet

Project: Husky Downward Camera Mount
Made by: Hari Nair
Area:
Subject:

No.
Date:
Checked by:
Date: Nov 12, 2026

Sheet 1 of 1

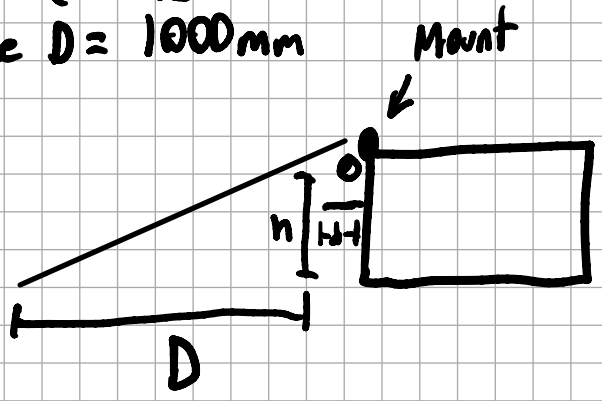
From sim:

lens $h = 375.5\,mm$

set back from bumper $d = 4.2\,mm$

RGB sensor for $\varphi = 42°$

Target distance $D = 1000\,mm$



$$\theta = \arctan\left(\frac{h}{D+d}\right) + \frac{\varphi}{2} = 20.5 + 21 = 41.5°$$

$$x_{far} = \frac{h}{\tan(\theta - \frac{\varphi}{2})} - d \approx 1000\,mm$$

$$x_{near} = \frac{h}{\tan(\theta + \frac{\varphi}{2})} - d = 191\,mm$$

∴ 41.5° tilt has 0.19 to 1m range

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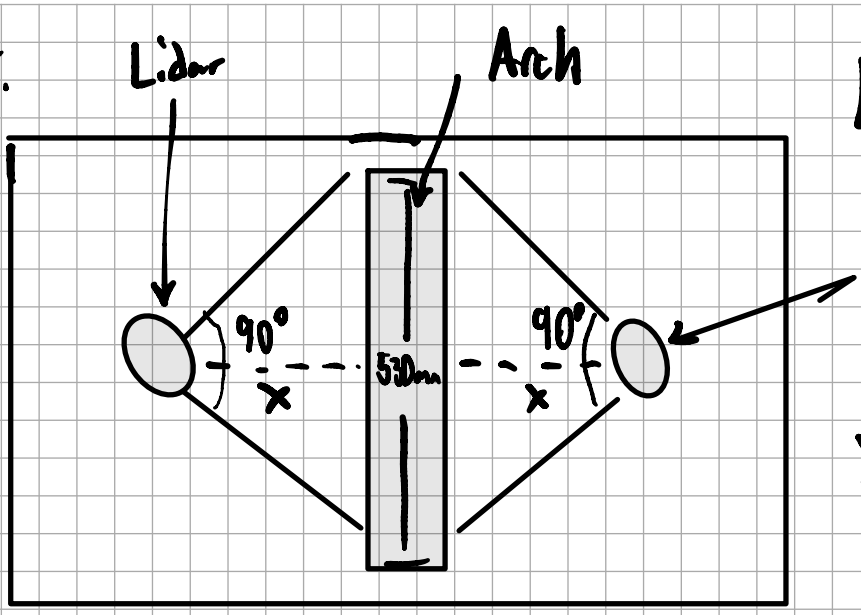
Calculation Sheet

Project: Husky Lidar Spacing
Made by: Hari Nair
Area:
Subject:

No.
Date:
Checked by:
Date: Nov 12, 2026

Sheet 1 of 1

Husky:



Lidar FOV: 270°

$X = \tan(45°) \cdot \frac{530}{2}$

$= 265mm$